

Attestia: A Decentralized Protocol for Verifiable Media Authenticity

Attestia

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Abstract

The rapid advancement of artificial intelligence has transformed how digital content is created and distributed—but at the cost of trust. AI-generated media, including deepfakes, is becoming indistinguishable from reality, undermining confidence in journalism, social platforms, and online information.

Attestia is a decentralized verification protocol designed to restore trust in digital media through verifiable and auditable authenticity. The system combines an AI-powered deepfake detector operated by Attestia with a decentralized network of independent attestors. Submitters request verification through the platform or API, while the Attestia detector produces an initial authenticity score and external attestors — running their own models — provide additional evaluations. These scores are aggregated into a consensus and recorded as on-chain attestations. To balance immediate usability with progressive decentralization, the weight of the Attestia detector is higher when few external attestors participate and progressively decreases as more independent attestations are submitted. The protocol aligns incentives through cryptoeconomics: attestors are rewarded when their evaluations align with consensus and penalized when they deviate. This creates a system where participation and accuracy are economically incentivized and low-quality or malicious behavior is discouraged. By combining AI detection, off-chain computation, and on-chain attestations, Attestia ensures scalability, privacy, and verifiability. By turning verification into an open and progressively decentralized market, Attestia establishes a new trust layer for the internet — where authenticity is not assumed, but proven.

Keywords

Blockchain, Web3, Deepfake, AI

1 Problem Statement

1.1 The Crisis of Digital Authenticity

The rapid proliferation of artificial intelligence has fundamentally altered the nature of digital content creation. Advanced generative models can now produce highly realistic images, videos, and audio that are often indistinguishable from authentic media. While these technologies unlock new creative possibilities, they simultaneously lead people scrolling their feeds to always ask themselves: “Am I looking at something real?” [4].

Deepfakes and synthetic media are no longer confined to research environments or niche applications. They are increasingly accessible, scalable, and deployable at low cost [8]. As a result, misinformation can be generated and disseminated

at unprecedented speed and scale, impacting journalism, public discourse, financial markets, and institutional trust [6]. In such an environment, the absence of reliable mechanisms to verify authenticity poses systemic risks to both digital ecosystems and society at large.

1.2 Limitations of Centralized Verification

Existing approaches to content verification rely heavily on centralized entities, such as fact-checking organizations, social media platforms, or specialized service providers (e.g., *Sensity AI*, *Hive AI*). In practice, these systems are predominantly based on proprietary automated models, with little to no human involvement in the verification loop. While these actors play an important role, their approaches exhibit several structural limitations [2],[5].

First, *verification processes are inherently opaque*. The underlying models, training data, and decision criteria are typically not disclosed, making it difficult for users to understand or independently validate how conclusions are reached.

Second, *algorithmic detection is not infallible*. Deepfake generation techniques evolve rapidly, and there exist multiple strategies to bypass detection models, including adversarial perturbations, compression artifacts, and distribution shifts. As a result, even state-of-the-art systems can produce unreliable or inconsistent outputs.

Third, *centralization introduces single points of failure*. A single provider ultimately determines the verification outcome, exposing the system to risks of bias, manipulation, censorship, or operational failure.

Finally, *verification outputs are not composable*. Results are typically confined within platform-specific infrastructures and cannot be easily reused, audited, or aggregated across different applications.

1.3 The Limitations of Purely Algorithmic Detection

Automated detection systems based on machine learning have emerged as a promising approach for identifying manipulated content at scale. These systems are capable of processing large volumes of media efficiently and detecting subtle statistical artifacts that may not be visible to human observers. As such, they represent a critical component in addressing the challenges posed by synthetic media [3].

However, relying exclusively on algorithmic detection introduces fundamental limitations. Many detection models operate as black boxes, offering limited transparency into how conclusions are reached and making their outputs difficult to interpret or audit. At the same time, the rapid evolution of generative models creates a persistent adversarial dynamic, where improvements in detection are continuously matched by more sophisticated forms of content synthesis. In addition, purely algorithmic approaches often struggle

to capture contextual and semantic aspects of authenticity, such as intent, narrative coherence, or domain-specific knowledge, which are essential in real-world verification scenarios. Finally, and most critically, the outputs of these systems are not inherently verifiable: users must ultimately trust the entity operating the model, rather than being able to independently validate the result [7]. An additional, often underestimated limitation lies in the data demands required to train effective detection models. Machine learning systems only achieve strong generalization when they are trained on large and diverse datasets. To reliably differentiate between authentic and synthetic media across many contexts, a model must be exposed to a wide range of examples during training. This diversity is not simply beneficial, but it is essential for robustness. Without sufficient coverage of rare, unusual, or out-of-distribution cases, even advanced detectors may fail, generating highly confident yet incorrect predictions when faced with unfamiliar inputs. Building and maintaining such comprehensive datasets is both expensive and complex to manage, while also introducing challenges related to annotation quality and potential bias. As a result, no single model or organization can realistically guarantee comprehensive and fully reliable detection across all types of content and domains.

While algorithmic detection is essential for scalability, it is insufficient on its own to establish trust in high-stakes environments. A robust solution must integrate automated analysis within a broader framework that enables complementary forms of evaluation and verifiable outcomes.

1.4 The Need for a Decentralized Verification Layer

The challenges outlined above point to the need for a new paradigm: a system where verification is not controlled by a single authority, but emerges from the coordinated input of multiple independent participants. Such a system should satisfy the following properties:

- **Decentralization:** No single entity should control the verification process.
- **Transparency:** Verification outcomes should be publicly auditable.
- **Incentive alignment:** Participants should be economically motivated to provide accurate evaluations.
- **Verifiability:** Results should be cryptographically provable and independently checkable.
- **Scalability:** The system should support large and diverse volumes of content and participants.

1.5 Toward Trustless Media Verification

Attestia is motivated by the need to establish a trustless, programmable layer for media authenticity. Instead of asking users to trust platforms, institutions, or proprietary models, the goal is to enable a system where authenticity is derived from verifiable attestations produced by a decentralized network. By

transforming verification into an open and incentive-driven process, Attestia addresses the fundamental limitations of existing approaches and lays the groundwork for a more resilient and trustworthy information ecosystem.

2 Core Concepts

2.1 Attestations

At the foundation of Attestia lies the concept of an *attestation*: a verifiable and auditable statement made by a participant about a specific piece of content. In the context of media authenticity, an attestation represents an evaluation of whether a given image, video, or other digital asset is authentic, manipulated, or synthetic with a certain confidence level. Attestations are cryptographically signed and linked to both the issuer and the subject of the evaluation. This ensures that each contribution is attributable, traceable, and cannot be altered after creation. Rather than producing a single authoritative verdict, the protocol aggregates multiple attestations, allowing authenticity to emerge as a consensus among independent evaluations.

To balance transparency, scalability, and privacy, Attestia adopts a hybrid approach to attestation storage. Individual evaluations are generated and stored off-chain, enabling efficient handling of large data and flexible access control. At the same time, aggregated results and their corresponding commitments are anchored on-chain, providing an immutable and publicly verifiable record. This design allows participants to independently verify outcomes without exposing sensitive intermediate data.

2.2 Participants

Attestia is built around a decentralized network of participants who contribute to the verification process through distinct but complementary roles.

Submitters provides media content to Attestia for its verification. These participants, which may include media organizations, platforms, or individual users.

Attestia Native Attester is an AI-powered deepfake detection system operated by Attestia itself. For each submitted media item, it produces an initial authenticity score that serves as a baseline input to the verification process. This score is not treated as a final authority, but as one contribution to the aggregate result. Its relative weight is higher when few independent attesters participate and progressively decreases as more external attestations are submitted, allowing the final outcome to increasingly reflect decentralized consensus as the network grows.

Independent Attesters are the core evaluators of the network. Each attester independently analyzes submitted content and produces an authenticity score, bringing to the network a diverse set of capabilities — ranging from advanced machine learning models to domain-specific human expertise and contextual reasoning. These different approaches are complementary and can collaborate to improve overall accuracy. Regardless of their nature, all attesters are required

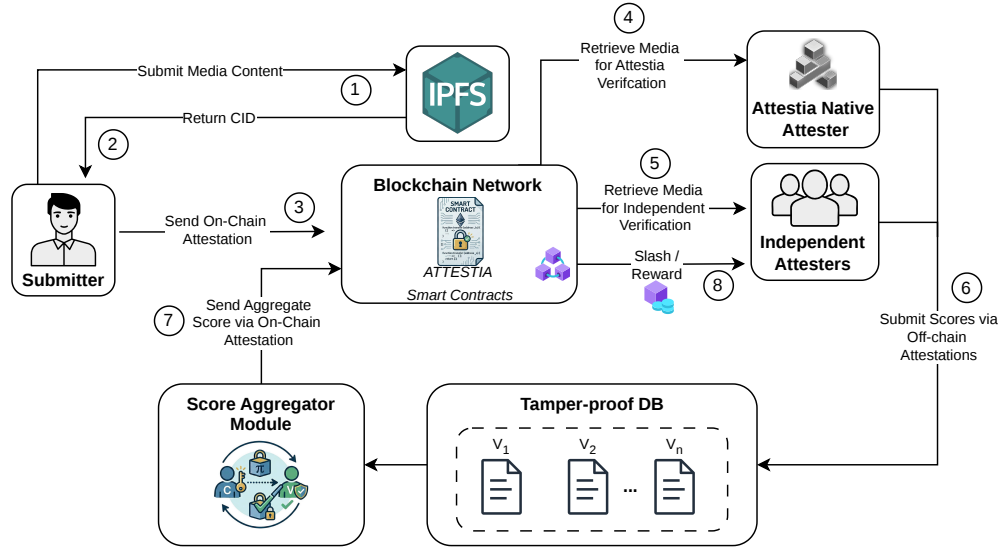


Figure 1: High-level architecture of Attestia

to stake tokens as a guarantee of honest behavior, operate under the same protocol rules, and are evaluated based on the quality and consistency of their contributions. Attesters whose assessments align with the network consensus are rewarded, while those who deviate significantly or behave maliciously are penalized through a slashing mechanism, losing a portion of their staked tokens.

In addition to submitters and attesters, the protocol can be consumed by external entities. Applications, platforms, and third-party services may query Attestia to retrieve authenticity attestations, integrating verifiable trust signals directly into their products and workflows.

2.3 Verification as a Market

Attestia models verification as an open and competitive market, where participants are incentivized to provide accurate assessments of content authenticity. Instead of relying on a predefined authority or a single detection system, the protocol enables multiple independent attesters to contribute their evaluations in parallel.

In this setting, different approaches to verification—ranging from advanced machine learning models to domain-specific human expertise—can coexist and compete. No single participant defines the ground truth; rather, authenticity emerges from the aggregation of diverse perspectives. This creates a dynamic environment in which better-performing attesters are naturally rewarded, while less reliable participants are penalized.

The market-based design fosters continuous improvement. As new detection techniques are developed and new participants enter the network, the overall quality of verification evolves over time. By aligning economic incentives with accuracy, Attestia transforms verification from a static process into

a self-improving system, capable of adapting to the rapidly changing landscape of synthetic media.

3 The Attestia Workflow

The Attestia protocol structures media verification as a time-bounded, multi-party process composed of sequential phases: submission, evaluation, aggregation, and settlement. This design enables open participation while preserving both confidentiality and verifiability. Figure 1 illustrates the overall architecture and the flow between these layers.

3.1 Submission Phase

The process is initiated by a *Submitter*, who submits a media item for verification. The media itself is not stored on-chain; instead, it is uploaded to a content-addressable storage system such as *IPFS*, ensuring both integrity and efficient distribution ①–②. An on-chain attestation — signed by Attestia — is then created via the *Ethereum Attestation Service* (EAS) [1] ③. This attestation encapsulates three elements: a cryptographic reference to the media (i.e., its *IPFS* hash), a minimal set of contextual metadata, and an expiration time defining the temporal boundary within which all attester evaluations must be submitted. By default, this expiration time is set to 30 minutes. The inclusion of an explicit expiration time is not merely operational—it defines the coordination window within which all evaluations must occur, effectively synchronizing participants under a shared constraint.

Once published, this attestation becomes the canonical entry point for the verification task. It is publicly discoverable, immutable, and uniquely identifiable, allowing downstream attestations to reference it unambiguously.

3.2 Verification Phase

Once the submission is created, the *Attestia Native Attester* analyzes the submitted media item and produces an initial authenticity score ④. This score acts as a baseline input to the verification process and enables the system to provide a meaningful assessment even when the number of independent attesters is limited. The native attester does not act as a final authority: its output is later combined with the evaluations submitted by external attesters according to the aggregation rule described in Section 3.4.

Independent Attesters continuously monitor the set of active submissions, either via the dashboard or via APIs ⑤. Each submission remains open until its expiration time, during which any eligible Attester may participate. Attesters may rely on machine learning models, heuristic analysis, or domain-specific expertise. What the protocol enforces is not *how* the evaluation is performed, but *when* and *under which information conditions* it is submitted. By requiring all scores to be provided within the predefined time window, *Attestia* reduces the risk of reactive or imitative behavior, preserving a degree of independence across contributions.

3.3 Confidential Off-Chain Attestations

Each evaluation made by the native and independent attesters is encoded as an off-chain attestation that references the original submission. These attestations are stored in a tamper-resistant data layer, where every record is bound to a cryptographic commitment ⑥. The decision to keep attester outputs off-chain is central to the protocol’s design. Publicly revealing intermediate scores would introduce undesirable dynamics—most notably herding effects, where later participants condition their outputs on earlier ones. By maintaining confidentiality until the aggregation phase, the system encourages genuine, independent assessments. At the same time, integrity is not sacrificed: each off-chain attestation is anchored on-chain via a timestamp, ensuring that once a score is submitted, its existence and timing are permanently recorded on the blockchain and cannot be altered without detection. This design also accommodates a heterogeneous set of attesters, including those operating proprietary models or handling sensitive signals, who may otherwise be unwilling to participate in a fully transparent environment.

3.4 Aggregation and Verifiable Computation

After the expiration time has elapsed, the system transitions from data collection to aggregation. An *Attestia Aggregator* module retrieves the score produced by the *Attestia Native Attester* together with all valid independent attester attestations and computes a consensus score according to a predefined weighted rule:

$$\bar{x} = w_A(N)x_A + (1 - w_A(N)) \frac{\sum_{m=1}^N \rho_m x_m}{\sum_{m=1}^N \rho_m} \quad (1)$$

where $x_A \in [0, 1]$ is the authenticity score produced by the *Attestia Native Attester*, $x_m \in [0, 1]$ is the authenticity score

submitted by independent attester m , N is the total number of participating independent attesters, and ρ_m is a reputation score reflecting the attester’s historical reliability within the network — its computation is discussed in detail in Section 4.3. The function $w_A(N) \in [0, 1]$ denotes the weight assigned to the native attester: it is higher when few independent attestations are available and decreases as more attesters participate, for example from $w_A(1) = 0.8$ to $w_A(N) = 0.2$ when $N > 5$.

While the aggregation function itself is straightforward, the challenge lies in making its execution trustless. To this end, the computation is accompanied by a *Zero-Knowledge (ZK) proof* attesting that the result was derived from the committed dataset, that all eligible inputs were considered, that the predefined weighting rule was applied correctly, and that the specified function was executed as intended. Crucially, this proof reveals nothing about individual contributions, exposing only the final aggregate.

3.5 On-Chain Settlement and Incentives

The final step consists in anchoring the result on-chain. The aggregated score, together with a commitment to the underlying proof, is recorded as an EAS attestation following a predefined schema ⑦. This attestation represents the protocol’s public output: a compact, tamper-resistant statement about the authenticity of the submitted media.

Once the result is finalized, the protocol evaluates the behavior of each Attester ⑧. Rather than rewarding absolute correctness—which is generally unknowable—the system measures alignment with the collective outcome. Attesters whose scores fall within an acceptable range of the aggregate are rewarded, while those that deviate significantly may incur penalties through a slashing mechanism. Over time, this feedback loop induces a form of emergent calibration: participants are incentivized to produce estimates that are both independent and statistically consistent with the broader set of evaluations. In this sense, consensus is not imposed ex ante, but arises ex post as the equilibrium of a cryptoeconomic process.

4 Cryptoeconomic Design

The security of *Attestia* emerges from a cryptoeconomic system that aligns incentives between submitters and attesters while adapting to the maturity of the network. The protocol is designed to balance three competing objectives: preventing spam and manipulation, incentivizing accurate verification, and enabling early-stage growth without penalizing honest participants. To achieve this, *Attestia* introduces differentiated staking requirements, phase-dependent rewards, confidence-aware slashing, and a reputation system that reinforces long-term reliability. In addition, the design ensures that the expected loss from dishonest behavior significantly exceeds the expected gain from a single verification task, making manipulation economically irrational.

The Attestia Detector participates in aggregation but does not receive attester rewards and is not subject to staking or slashing. Its function is operational and commercial: it provides a reliable baseline signal, supports early-stage network bootstrapping, and improves response time for submitters.

4.1 Staking Model

Attesters stake tokens to participate in the evaluation process. Their stake acts as active collateral: it grants access to rewards but is also subject to slashing. Since attesters directly determine the credibility of the protocol, their economic exposure is significantly higher. At initialization, the protocol adopts a default attester stake of:

$$s_o = 0.10 \text{ ETH.} \quad (2)$$

To maintain a consistent level of economic security, the stakes are defined in ETH-equivalent terms and dynamically adjusted using price oracles. This prevents fluctuations in the native token price from weakening the protocol’s security guarantees and ensures that the economic exposure of each participant remains meaningful regardless of market conditions.

4.2 Reward Mechanism

Rewards are composed of a stable component denominated in ETH and a variable component denominated in the native token (introduced in Section 5). Each attester m submits a score x_m for a given content item. The deviation from consensus is defined as:

$$d_m = |x_m - \bar{x}|. \quad (3)$$

Rewards are composed of two components: alignment with consensus and contribution to the informativeness of the aggregate outcome. The reward function is defined as:

$$r_m = R_\phi \cdot (w_1 e^{-\alpha d_m} + w_2 I_m) \cdot \rho_m, \quad (4)$$

where R_ϕ is the phase-dependent reward pool, α controls sensitivity to deviation, ρ_m is the attester’s reputation multiplier, and I_m is an influence factor measuring the marginal contribution of attester m to the aggregate result (e.g., the change in $\bar{x}$ when excluding m). The weights satisfy $w_1 + w_2 = 1$.

This formulation ensures that attesters are incentivized not only to align with consensus, but also to provide independent and informative evaluations. Small deviations result in marginal reward reductions, while large deviations are penalized exponentially. At the same time, attesters whose contributions meaningfully affect the outcome are rewarded, discouraging low-effort or purely imitative strategies.

4.3 Reputation System

Each attester maintains a reputation score ρ_m reflecting their historical performance. Reputation evolves over time as an exponentially weighted average of past alignment scores:

$$\rho_m^{(t)} = \lambda \rho_m^{(t-1)} + (1 - \lambda) e^{-\alpha d_m}, \quad (5)$$

with $\lambda = 0.8$.

This formulation gives more weight to recent performance while still retaining memory of past behavior. Reputation is bounded within:

$$\rho_m \in [0.5, 1.5]. \quad (6)$$

Reputation acts as a soft multiplier on rewards without affecting the aggregation mechanism, ensuring that high-performing attesters are incentivized while preserving openness and preventing centralization of influence.

4.4 Slashing Mechanism

Slashing is designed to penalize clear outliers while preserving tolerance for honest disagreement. The key quantity governing slashing is the deviation value d_m , which measures how far a attester’s score departs from the aggregate defined in Section 4.2.

However, a large deviation alone is not sufficient to trigger a penalty. A attester may legitimately disagree with the majority in situations where the network is uncertain—for example, when submitted scores are widely dispersed. In such cases, high variance among attester scores indicates that the content may be ambiguous or that consensus has not yet emerged. Penalizing deviations under these conditions would be economically unjustified and could discourage participation on genuinely difficult evaluations.

For this reason, slashing is additionally conditioned on the variance of submitted scores, denoted σ^2 . Low variance signals that the attester set has reached a strong consensus, making a large individual deviation less defensible as honest disagreement. Accordingly, slashing is triggered only when both of the following conditions are satisfied:

$$d_m > \delta \quad \text{and} \quad \sigma^2 < \tau \quad (7)$$

where δ is the deviation threshold and τ is the variance ceiling above which slashing is suppressed. When both conditions are met, a fraction of the attester’s stake is forfeited.

The precise values of δ , τ , and the penalty fraction are phase-dependent and specified in Section 4.5.

4.5 Network Phases

The incentive structure of Attestia evolves as the independent attester set grows. In early stages, consensus is fragile and penalties must be conservative. The Attestia Native Attester mitigates this cold-start problem by providing an initial baseline score whose weight decreases as more independent attesters participate. As the network matures, stronger economic guarantees can be enforced. Let N be the number of active independent attesters. The protocol operates under three phases.

Phase 0: Bootstrapping ($N < 5$). At this stage, the independent attester set is too small to produce a reliable consensus on its own. The Attestia Native Attester therefore carries a higher weight in the aggregate score, allowing the protocol to provide meaningful verification results while the

network is still growing. Disagreement between attesters is expected and does not necessarily indicate malicious behavior. Slashing is disabled in this phase. Attesters receive rewards at 50% of the base reward, reflecting the lower confidence in the decentralized component of the aggregate outcome. Reputation updates remain active, allowing the protocol to start building a performance history. The primary objective of this phase is to maximize participation and bootstrap the attester set.

Phase 1: Weak Consensus ($5 \leq N \leq 20$). The network begins to produce meaningful decentralized signals, while the relative influence of the Attestia Native Attester decreases according to the predefined weighting rule. Slashing is enabled selectively: a penalty equal to 10% of the attester stake is applied only when $d_m > 0.25$ and $\sigma^2 < 0.015$. This joint condition ensures that penalties are triggered only when the network has converged with sufficient confidence. Rewards are distributed at 80% of the base rate.

Phase 2: Mature Network ($N > 20$). The aggregated outcome is primarily driven by the independent attester network, while the Attestia Native Attester remains a minority baseline signal. At this stage, the variance condition is lifted. Slashing is applied whenever $d_m > 0.20$, with a penalty equal to 25% of the attester stake. Rewards are distributed at full rate.

5 Token Design

5.1 Token Utility

The native token of the protocol, **ATTA**, serves as the economic coordination layer of Attestia. Its role is not to act as a general-purpose currency, but to secure the behavior of attesters and enable trustless coordination within the network.

The protocol distinguishes between two categories of participants with different relationships to ATTA.

Submitters and consumers of attestations — media organizations, platforms, and downstream applications — interact with the protocol exclusively through standard payment rails, using ETH. They are not required to acquire or manage ATTA at any point. From their perspective, the token is entirely abstracted from the user experience.

Attesters, by contrast, are the primary economic actors within the ATTA ecosystem. Their relationship with the token evolves across two distinct stages of the protocol, as described in Section 5.3. In the mature network, attesters are required to lock ATTA as collateral in order to participate in the evaluation process. This requirement creates structural demand for the token and aligns the economic interests of attesters with the long-term health of the network: a attester who holds staked ATTA has a direct financial stake in the credibility of the protocol.

At the protocol level, ATTA fulfills two concrete functions. First, it enables stake-backed participation in the mature network: once the token has established a stable market price, attesters are required to lock ATTA as collateral. Second,

it serves as a long-term incentive mechanism: attesters accumulate ATTA as the variable component of their reward from the earliest stages of the network, aligning their interests with the growth of the protocol before the staking transition occurs.

5.2 Reward Denomination Model

A critical design decision in any token-based incentive system concerns the denomination of rewards: whether participants are compensated in the native token, in stable assets, or in a combination of both.

Compensating attesters exclusively in ATTA would tie their real income entirely to the market price of the token. In practice, an exclusively token-denominated reward structure would deter technically capable attesters and attract primarily speculative participants, undermining the credibility of the consensus mechanism.

For this reason, Attestia adopts a mixed denomination model in which attester rewards are composed of two components. The first is a stable component, denominated in ETH, derived directly from the fee paid by submitters. This component guarantees a minimum level of real economic return for attesters regardless of token price conditions, making participation viable for operators with concrete infrastructure costs. The second is a variable component, denominated in ATTA and funded through protocol emissions. This component aligns the long-term interests of attesters with the success of the protocol: as ATTA appreciates with adoption, so does the value of their accumulated rewards.

The proportion between the two components is not fixed and evolves in a way that reflects the maturity of both the protocol and the token. In the initial phase, the stable component is dominant: attesters must be able to cover real operational costs today, and cannot rely on an asset that has not yet established a market price. The variable ATTA component is present but in a minority, serving as a long-term incentive for attesters who believe in the protocol and are willing to accumulate the token ahead of its market formation. As the protocol enters the mature phase, emissions decrease and ATTA has established a stable market price. The stable component remains significant, sustained by the growth of fee income, while the ATTA component becomes smaller in quantity but not necessarily in value — since each unit of ATTA is now priced by an active market.

5.3 Staking Transition Model

A fundamental challenge in launching a token-based protocol is the cold-start problem for the token itself: attesters cannot be expected to stake an asset that has no established market value, yet the token cannot acquire market value without an active attester set that creates demand for it. Requiring ATTA staking from day one would make it impossible to attract the first participants, regardless of the protocol's technical merits. To address this, Attestia adopts a two-phase staking model that separates the bootstrapping of the attester network from

the bootstrapping of the token.

In the initial phase, attesters stake ETH as collateral. This allows participation to begin immediately, without any dependency on ATTA's market price. ETH is a liquid, widely held asset with well-understood value, removing the barrier to entry for technically capable attesters who would otherwise be unwilling to acquire and lock an illiquid or unpriced token. During this phase, attesters are compensated through the mixed denomination model described in Section 5.2: a stable component in ETH derived from submitter fees, and a variable component in ATTA distributed through protocol emissions. Over time, attesters accumulate ATTA as part of their rewards without being required to purchase it on the open market.

The transition to the mature phase occurs once ATTA has established a stable and liquid market price. At this point, the staking requirement migrates from ETH to ATTA. Attesters who have accumulated sufficient ATTA through participation can transition their stake without any additional capital outlay, while new entrants must acquire ATTA on the open market. This migration creates the structural demand for the token that underpins its long-term value: every new attester entering the network represents incremental buy pressure on ATTA, aligning token appreciation with protocol growth.

5.4 Emission Model

The total supply of ATTA is fixed at launch, establishing a hard cap that provides resistance to long-term inflation and gives participants a well-defined dilution horizon. Within this fixed supply, emissions follow an *exponentially decaying schedule*, rather than a constant issuance rate, ensuring that inflation is front-loaded and progressively reduced over time.

Formally, the emission rate is defined as a function of network progression:

$$E(N) = E_0 \cdot e^{-\lambda N}$$

where E_0 is the initial emission rate, $\lambda > 0$ is the decay constant, and N denotes the network phase as defined in Section 4.5. This formulation ensures a smooth and continuous reduction in emissions as the protocol matures.

During the bootstrapping phase ($N < 5$), the emission rate is at or near its maximum. Fee income is negligible and the attester set is still forming, so the protocol relies primarily on token issuance to incentivize participation. Emissions in this phase are directed exclusively to attesters, rewarding early involvement and enabling the accumulation of reputation history before organic demand emerges.

As the network enters the expansion phase ($5 \leq N \leq 20$), emissions decline according to the exponential schedule. At the same time, fee income begins to contribute meaningfully to the reward pool, reducing the protocol's reliance on inflation. Token emissions continue to supplement rewards, but their relative importance decreases as network usage grows.

In the maturity phase ($N > 20$), the emission rate approaches a low asymptotic level. At this stage, the protocol is sustained predominantly by fee-driven rewards, while residual emissions provide a baseline incentive for attesters and help maintain

liquidity during periods of low activity, without introducing significant dilution.

To counterbalance new issuance across all phases, the protocol incorporates a burn mechanism: a fixed fraction of the fee paid by each submitter is permanently removed from circulation. This establishes a direct link between network activity and token scarcity—higher usage leads to higher burn rates. When demand is low, emissions dominate and net supply expands. As adoption increases, the burn rate progressively offsets issuance. Under sufficiently high demand, the burn rate may exceed emissions, rendering ATTA net deflationary. This dynamic anchors token value to actual protocol usage rather than purely speculative demand.

6 Use Cases

Attestia is designed as a general-purpose verification layer for digital content, enabling trust to emerge from decentralized evaluation rather than centralized authority. Its applicability spans multiple domains where authenticity, provenance, and credibility are critical. The following use cases illustrate how the protocol can be integrated across different contexts.

6.1 Media & Journalism

In journalism, the erosion of trust is no longer driven solely by misinformation, but by the rapid proliferation of AI-generated and AI-manipulated content. Advances in generative models have made it possible to produce highly realistic images, videos, and audio at near-zero cost, often indistinguishable from authentic media even under expert inspection. This introduces systemic risks: fabricated evidence can be used to manipulate public opinion, synthetic interviews can be presented as real, and authentic content can be dismissed as fake, giving rise to the so-called "liar's dividend." Traditional verification workflows rely on editorial processes that are opaque, slow, and difficult to scale, making them ill-suited for a landscape where content can be generated and disseminated globally within seconds.

Attestia introduces a verifiable authenticity layer for media organizations, enabling newsrooms to attach cryptographic attestations to images, videos, and reports. Independent attesters — ranging from AI systems to specialized analysts — evaluate content and contribute authenticity scores that are aggregated into a public, tamper-resistant signal.

This allows publishers to embed verifiable trust directly into their content distribution pipelines, while readers and downstream platforms can independently validate the integrity of the information they consume. Rather than trusting a single newsroom, users rely on a transparent, multi-party verification process.

6.2 Social Platforms

Social platforms operate at a scale where manual moderation is fundamentally insufficient, particularly in the presence of AI-generated content that can be produced in massive volumes and tailored for virality. Synthetic media can be deployed

to amplify disinformation campaigns, create coordinated narratives, or bypass traditional detection systems through rapid iteration. At the same time, centralized moderation systems introduce concerns around bias, censorship, and lack of accountability, especially when opaque algorithms determine the visibility or suppression of content.

Attestia enables platforms to externalize content verification into an open market of attesters. Content can be programmatically routed through the protocol, generating authenticity attestations that inform ranking algorithms, visibility decisions, or warning labels. This architecture decouples content distribution from content verification. Platforms retain control over user experience, while verification becomes a composable, interoperable primitive. Importantly, the system introduces economic accountability: attesters are incentivized to provide accurate assessments, and penalized for systematic misalignment.

As a result, moderation evolves from a purely reactive and policy-driven process into a hybrid system combining automated detection, decentralized evaluation, and market-based incentives, better suited to counter the scale and adaptability of AI-generated content.

6.3 Financial / Legal Evidence

In financial and legal contexts, the integrity of digital evidence is paramount. Documents, records, and multimedia artifacts often require verification processes that are both auditable and resistant to tampering.

Attestia provides a decentralized mechanism to certify the authenticity and integrity of such evidence. Multiple independent attesters can assess documents or datasets, producing attestations that are aggregated into a verifiable outcome anchored on-chain. This approach enhances traditional verification workflows by introducing transparency and redundancy. Rather than relying on a single certifying authority, trust is distributed across a network of evaluators, each economically incentivized to act honestly.

Potential applications include due diligence processes, forensic analysis of digital evidence, verification of financial disclosures, and certification of real-world assets. In all cases, Attestia enables a shift from trust-by-authority to trust-by-verification.

7 Conclusion

The rapid advancement of generative AI has fundamentally challenged the reliability of digital information, creating an urgent need for systems that can restore trust in online content. Attestia addresses this challenge by introducing a verifiable and incentive-aligned infrastructure for media authenticity. By combining the Attestia Native Attester with a decentralized network of independent attesters, the system provides useful verification results from the earliest stages of adoption while progressively shifting trust toward decentralized consensus as participation grows.

The native attester produces an initial baseline score for each submitted media item, improving responsiveness and reducing

the cold-start limitations of purely decentralized verification. As more independent attesters contribute, their evaluations increasingly determine the final aggregate outcome, which is recorded as an on-chain attestation. The integration of staking, reputation, rewards, and slashing mechanisms ensures that accurate behavior is economically incentivized, while low-quality or malicious contributions are discouraged. As synthetic media continues to evolve, Attestia provides a flexible and extensible foundation for verifiable media authenticity, enabling applications across journalism, social platforms, and high-stakes digital evidence. In doing so, it contributes to the development of a more transparent, resilient, and trustworthy information ecosystem, where authenticity is not assumed or dictated by a single authority, but evaluated, aggregated, and proven.

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